MIU: model notes

Based on `MIU.mod`; Walsh (2010), Eq. 2.64 and Appendix 2.7, p. 85.

Underlying preferences

$$S = ac^{1-\eta_m} + (1-a)m^{1-\eta_m}, \qquad U = \frac{S^{(1-\eta_c)/(1-\eta_m)}}{1-\eta_c} + \Psi\frac{(1-n)^{1-\eta_n}}{1-\eta_n}.$$

Walsh's Φ, b, η map to η_c, η_m, η_n. At $\eta_n = 1$, use $\Psi\log(1-n)$. The weight a is on consumption. Money affects marginal consumption utility unless $\eta_m = \eta_c$.

$$MC = \left[\frac{a}{1-a}\frac{i^{ss}}{1+i^{ss}}\right]^{-1/\eta_m}, \qquad \gamma = \left[1 + \frac{1-a}{a}MC^{1-\eta_m}\right]^{-1},$$
$$\omega_1 = \gamma\eta_c + (1-\gamma)\eta_m, \qquad \omega_2 = (\eta_m - \eta_c)(1-\gamma).$$

Here $MC = m^{ss}/c^{ss}$, not marginal cost. Walsh's $\Omega_1 = -\omega_1$ and $\Omega_2 = \omega_2$.

Scope and calibration

These are stationary, log-linear business-cycle equations. Quantities are log deviations; r, i and π are level deviations of net rates. The system uses small-rate approximations, not exact gross-rate scaling. The leisure curvature implies a Frisch elasticity $(1-n^{ss})/(\eta_n n^{ss})$.

The code fixes `nss`; Ψ is implicit. Changing trend inflation while retaining `nss` changes the implied leisure weight. It does not trace steady-state hours with preferences fixed. Additive leisure preferences also generally do not support constant hours along balanced growth unless consumption-composite curvature is one; these exercises abstract from growth.

Linear equations implemented in the code

All equations are conditional on information at t; future variables in Dynare imply conditional expectations. Hats are suppressed, as in the code. Write $YK = y^{ss}/k^{ss}$, $CK = YK - \delta$ and $NL = n^{ss}/(1 - n^{ss})$.

$$
\begin{aligned}
y_t &= z_t + \alpha k_{t-1} + (1-\alpha) n_t, \\
YK\, y_t &= CK\, c_t + \delta x_t, \\
k_t &= (1-\delta) k_{t-1} + \delta x_t, \\
r_t &= \alpha YK (E_t y_{t+1} - k_t), \\
\omega_1 c_t &= \omega_1 E_t c_{t+1} - r_t - \omega_2 (E_t m_{t+1} - m_t), \\
y_t - \omega_1 c_t + \omega_2 m_t &= (1 + \eta_n NL) n_t, \\
i_t &= r_t + E_t \pi_{t+1}, \\
m_t &= c_t - \frac{i_t}{\eta_m i^{ss}}, \\
m_t &= m_{t-1} - \pi_t + \mu_t, \\
z_t &= \rho_z z_{t-1} + \epsilon_{z,t}, \\
\mu_t &= \rho_\mu \mu_{t-1} + \phi_z z_{t-1} + \epsilon_{\mu,t}.
\end{aligned}
$$

In code: $\alpha =$ `alppha`, $\beta =$ `betta`, $i =$ `int`, $\pi =$ `pii`, investment is `x`. The current calibration sets $\phi_z = 0$.